Coding For Placement Series

Capstone Problem Bank

100 problem statements for the Java, Python and C tracks

STUDENT HANDOUT

BIZOTIC

Empowering students towards their first job!

v1

How the Capstone Works

You will build one working console application in your track language (Java, Python or C), alone or with one partner. You choose one problem from this bank in the capstone launch session in Phase 1, build it on your own before Phase 2, and present it in the capstone review session in Phase 2.

Why this matters: in almost every technical interview you will be asked, "Tell me about a project you have built." After this capstone you will have a program you wrote yourself, can run in front of anyone, and can explain.

The rules

Rule	Detail
Team size	Maximum 2 students per team. Working alone is fully acceptable. Teams of 3 are not allowed.
Problem choice	Choose one problem from this bank (CP-001 to CP-100). Own ideas are not accepted in this cycle.
No duplicates	No two teams in the same batch should take the same problem. This is your responsibility: discuss with your classmates and make sure your problem is not already taken.
Language and type	Your track language (Java, Python or C). A menu-driven console application. No GUI, website or database is needed. Saving to a text or CSV file is allowed as a stretch feature.
Scope	All 5 Core Features are compulsory. Stretch Features are optional.
Use of AI	You may use AI tools as a coach: to explain an error, review your code or suggest test cases. AI must not write your project. In the review, you will be asked a question on your own code.
Building	You build the project on your own between Phase 1 and Phase 2. There are no checkpoints in between.
What to bring	Your working project, ready to run, and your GitHub repository link. The review happens during the Phase 2 training and can be in any session, so have it ready from the first morning of Phase 2.

How you will be marked (10 marks)

Parameter	Marks	What earns full marks
It works	5	The program runs, and the features you are asked to show work correctly, including wrong input
Viva	3	You answer the trainer's question on your own code clearly and correctly
Demo	2	A clear, confident 2-minute demo

Each team gets **3 minutes** in the review: a 2-minute demo, then 1 minute for a question and feedback. In a team of two, the trainer chooses who answers, so both of you must know the whole code.

Suggested plan for building

When	What you should have done
End of the launch session	Team formed, problem chosen, Project Plan Card checked by the trainer, GitHub repository created
Week 1	Data design done, main menu running, Core Feature 1 working
Week 2	Core Features 1 and 2 working
Week 3	Core Features 3, 4 and 5 working, wrong input handled
Week 4	Tested with your edge cases, README written, demo practised
Review session (Phase 2)	Bring the working project and your GitHub link

How to choose your problem

- **Pick one you can finish.** A finished Foundation project beats an unfinished Advanced one, in marks and in interviews.
- **Working alone?** Choose Foundation or Intermediate.
- **Choose a domain you would enjoy talking about.** If you want a banking or fintech job, a banking problem makes a better interview story.
- **Read all 5 Core Features before choosing.** If you cannot picture how to build any one of them, choose another problem.
- **Everything here can be built with what you learnt in Phase 1**: conditionals, loops, methods or functions, arrays or lists, and strings. Classes, objects and collections from Phase 2 are an upgrade, not a requirement.

Level	What it means	Problems in the bank
Foundation	Straightforward records and rules. Good for solo students.	40
Intermediate	More rules, more records, some calculation or matching logic.	46
Advanced	Needs an algorithmic idea such as scheduling, routing, simulation or similarity.	14

Problem Index

ID	Problem	Domain	Level
CP-001	VaultCore: Core Banking Transaction Engine	Banking and FinTech	Foundation
CP-002	SpendSense: Smart Budget Monitoring System with Threshold Alerts	Banking and FinTech	Foundation
CP-003	LoanLens: EMI Amortisation and Loan Comparison Engine	Banking and FinTech	Foundation
CP-004	FraudShield: Rule-Based UPI Transaction Risk Engine	Banking and FinTech	Intermediate
CP-005	CreditPulse: Explainable Credit Score Estimation Engine	Banking and FinTech	Intermediate
CP-006	PortfolioIQ: Equity Portfolio and P&L Analytics System	Banking and FinTech	Intermediate
CP-007	EvenSplit: Group Expense Settlement Optimiser	Banking and FinTech	Intermediate
CP-008	CashFlowX: ATM Cash Dispensing and Security Simulator	Banking and FinTech	Foundation
CP-009	GoalVault: Goal-Based Savings and SIP Planning Engine	Banking and FinTech	Foundation
CP-010	TaxLedger: GST-Compliant Invoicing and Tax Engine	Banking and FinTech	Intermediate
CP-011	SwiftCart: Point-of-Sale Billing and Inventory System	E-Commerce and Retail	Foundation
CP-012	StockSense: Predictive Inventory Reorder Engine	E-Commerce and Retail	Intermediate
CP-013	BuyNext: Co-Purchase Product Recommendation Engine	E-Commerce and Retail	Advanced
CP-014	DealForge: Coupon Validation and Best-Offer Optimisation Engine	E-Commerce and Retail	Intermediate
CP-015	RushQueue: Fair-Allocation Flash Sale Order Engine	E-Commerce and Retail	Advanced
CP-016	ReturnFlow: Policy-Driven Returns and Refund Management System	E-Commerce and Retail	Foundation
CP-017	TrackPoint: Order Lifecycle and Delivery Tracking System	E-Commerce and Retail	Foundation
CP-018	PriceRadar: Multi-Seller Price Comparison and Alert Engine	E-Commerce and Retail	Intermediate
CP-019	LoyaltyLoop: Tiered Customer Rewards Management System	E-Commerce and Retail	Foundation
CP-020	SalesScope: Retail Sales Analytics Dashboard	E-Commerce and Retail	Intermediate
CP-021	MediQueue: Priority-Based Patient Queue Management System	Healthcare	Foundation

ID	Problem	Domain	Level
CP-022	PharmaTrack: Batch-Level Medicine Inventory and Expiry Control System	Healthcare	Foundation
CP-023	LifeLink: Blood Inventory and Donor Matching System	Healthcare	Intermediate
CP-024	VitalWatch: Patient Vitals Monitoring and Early-Alert System	Healthcare	Intermediate
CP-025	BedFlow: Hospital Bed Allocation and Occupancy Management System	Healthcare	Intermediate
CP-026	NutriPlan: Personalised Calorie and Macro Planning Engine	Healthcare	Foundation
CP-027	DoseMate: Medication Scheduling and Adherence Tracker	Healthcare	Foundation
CP-028	LabInsight: Diagnostic Report Analysis and Trend Engine	Healthcare	Intermediate
CP-029	ClaimSure: Health Insurance Claim Adjudication Engine	Healthcare	Intermediate
CP-030	OutbreakSim: Grid-Based Epidemic Spread Simulation Model	Healthcare	Advanced
CP-031	GradeCore: Academic Result Processing and Ranking Engine	Education and Campus	Foundation
CP-032	AttendIQ: Attendance Analytics and Shortage Prediction System	Education and Campus	Foundation
CP-033	QuizForge: Randomised Assessment and Leaderboard Engine	Education and Campus	Intermediate
CP-034	ClashGuard: Timetable Conflict Detection and Resolution Engine	Education and Campus	Advanced
CP-035	ShelfWise: Library Circulation and Reservation Management System	Education and Campus	Foundation
CP-036	SeatPlanner: Constraint-Based Exam Seating Allocation Engine	Education and Campus	Intermediate
CP-037	RoomRight: Priority-Based Hostel Allocation System	Education and Campus	Intermediate
CP-038	PlaceMatch: Multi-Criteria Placement Eligibility Engine	Education and Campus	Intermediate
CP-039	FeedbackIQ: Course Feedback Sentiment and Ratings Analyser	Education and Campus	Intermediate
CP-040	RecallPro: Spaced-Repetition Learning Scheduler	Education and Campus	Advanced
CP-041	RouteSeat: Intercity Bus Reservation and Seat Management System	Travel, Transport and Logistics	Foundation
CP-042	MetroMind: Shortest-Route and Fare Computation Engine	Travel, Transport and Logistics	Advanced
CP-043	ParkSmart: Multi-Level Parking Allocation and Billing System	Travel, Transport and Logistics	Foundation

ID	Problem	Domain	Level
CP-044	RideFare: Dynamic Surge Pricing Engine for Ride-Hailing	Travel, Transport and Logistics	Intermediate
CP-045	ShipTrail: Zone-Based Courier Tariff and Shipment Tracking System	Travel, Transport and Logistics	Intermediate
CP-046	SkyBook: Dynamic-Fare Flight Reservation and Waitlist Engine	Travel, Transport and Logistics	Intermediate
CP-047	FleetPulse: Fleet Fuel Efficiency and Maintenance Analytics System	Travel, Transport and Logistics	Foundation
CP-048	LoadIQ: Priority-Driven Delivery Load Optimisation Engine	Travel, Transport and Logistics	Advanced
CP-049	PlatformPro: Interval-Scheduling Platform Allocation Engine	Travel, Transport and Logistics	Advanced
CP-050	TripCraft: Itinerary Scheduling and Budget Control Planner	Travel, Transport and Logistics	Foundation
CP-051	LeaveFlow: Leave Approval Workflow and Team Availability System	HR and Workplace	Foundation
CP-052	TalentTrack: Skill-Matching Applicant Tracking System	HR and Workplace	Intermediate
CP-053	ShiftGenie: Constraint-Based Workforce Roster Generator	HR and Workplace	Advanced
CP-054	PunchLog: Rule-Based Attendance and Overtime Engine	HR and Workplace	Foundation
CP-055	RoomSync: Conflict-Free Meeting Room Booking System	HR and Workplace	Foundation
CP-056	SkillGrid: Skill-Gap Analysis and Training Recommendation Engine	HR and Workplace	Intermediate
CP-057	ClaimFlow: Multi-Level Expense Approval Workflow Engine	HR and Workplace	Intermediate
CP-058	AssetTrail: IT Asset Lifecycle and Allocation Tracker	HR and Workplace	Foundation
CP-059	GatePass: Visitor Access and Security Management System	HR and Workplace	Foundation
CP-060	ResolveDesk: SLA-Driven Helpdesk Ticketing System	HR and Workplace	Intermediate
CP-061	KitchenSync: Restaurant Order Routing and Kitchen Display System	Food, Hospitality and Events	Intermediate
CP-062	StayEase: Hotel Reservation and Availability Engine	Food, Hospitality and Events	Foundation
CP-063	DashDispatch: Nearest-Rider Delivery Assignment Engine	Food, Hospitality and Events	Advanced
CP-064	TicketVerse: Seat-Category Event Ticketing System	Food, Hospitality and Events	Foundation
CP-065	CookScale: Recipe Scaling and Smart Grocery Planner	Food, Hospitality and Events	Foundation

ID	Problem	Domain	Level
CP-066	QuickBite: Slot-Based Canteen Pre-Ordering System	Food, Hospitality and Events	Foundation
CP-067	QuoteChef: Catering Cost Estimation and Quotation Engine	Food, Hospitality and Events	Foundation
CP-068	GuestGrid: Event Guest and Table Seating Manager	Food, Hospitality and Events	Foundation
CP-069	TableTurn: Best-Fit Table Reservation and Waitlist Engine	Food, Hospitality and Events	Intermediate
CP-070	EventLedger: Event Budget and Vendor Payment Tracker	Food, Hospitality and Events	Intermediate
CP-071	PowerBill: Slab-Based Utility Billing Engine	Public Services, Utilities and Smart City	Foundation
CP-072	AquaRoute: Priority-Based Water Supply Scheduling System	Public Services, Utilities and Smart City	Intermediate
CP-073	CivicConnect: Complaint Routing and Escalation System	Public Services, Utilities and Smart City	Intermediate
CP-074	SignalSense: Adaptive Traffic Signal Simulation Model	Public Services, Utilities and Smart City	Advanced
CP-075	BinSense: Smart Waste Monitoring and Collection Route Planner	Public Services, Utilities and Smart City	Intermediate
CP-076	ChallanTrack: Traffic Violation and Penalty Management System	Public Services, Utilities and Smart City	Foundation
CP-077	RationRight: Entitlement-Based Public Distribution Tracker	Public Services, Utilities and Smart City	Intermediate
CP-078	GasGo: LPG Refill Booking and Delivery Management System	Public Services, Utilities and Smart City	Foundation
CP-079	VoteSecure: Polling Booth Verification and Vote Counting Simulator	Public Services, Utilities and Smart City	Intermediate
CP-080	TaxAssess: Multi-Factor Property Tax Computation Engine	Public Services, Utilities and Smart City	Foundation
CP-081	CricStats: Ball-by-Ball Cricket Scoring and Analytics Engine	Media, Sports and Entertainment	Intermediate
CP-082	FantasyXI: Constraint-Validated Fantasy Team Builder and Points Engine	Media, Sports and Entertainment	Advanced
CP-083	ScreenSeat: Cinema Show and Seat Booking System	Media, Sports and Entertainment	Foundation
CP-084	BeatQueue: Playlist Management and Smart Shuffle Engine	Media, Sports and Entertainment	Foundation
CP-085	FixtureGen: Round-Robin Tournament Scheduling Engine	Media, Sports and Entertainment	Intermediate
CP-086	FitTrack: Gym Membership and Access Management System	Media, Sports and Entertainment	Foundation
CP-087	WatchNext: Genre-Preference Content Recommendation Engine	Media, Sports and Entertainment	Intermediate

ID	Problem	Domain	Level
CP-088	AthleteIQ: Athlete Performance Tracking and Selection System	Media, Sports and Entertainment	Intermediate
CP-089	PostPilot: Multi-Platform Social Media Scheduling Engine	Media, Sports and Entertainment	Intermediate
CP-090	TuneRecap: Music Listening Analytics and Year-in-Review Generator	Media, Sports and Entertainment	Intermediate
CP-091	CipherVault: Password Strength Analyser and Credential Vault	Productivity, Security and Developer Tools	Intermediate
CP-092	PriorityPro: Eisenhower-Matrix Task Prioritisation Engine	Productivity, Security and Developer Tools	Foundation
CP-093	LogLens: Server Log Analytics and Anomaly Detection Tool	Productivity, Security and Developer Tools	Intermediate
CP-094	CopyCheck: Text Similarity and Plagiarism Detection Engine	Productivity, Security and Developer Tools	Advanced
CP-095	BugBoard: Defect Lifecycle Tracking System	Productivity, Security and Developer Tools	Intermediate
CP-096	DiffTrack: Document Versioning and Diff Engine	Productivity, Security and Developer Tools	Advanced
CP-097	ShortLink: Base62 URL Shortening and Click Analytics Service	Productivity, Security and Developer Tools	Intermediate
CP-098	FocusFlow: Study Session Analytics and Streak Tracker	Productivity, Security and Developer Tools	Foundation
CP-099	CleanSlate: Data Validation and Cleansing Utility	Productivity, Security and Developer Tools	Intermediate
CP-100	AskCampus: Rule-Based College FAQ Chatbot	Productivity, Security and Developer Tools	Intermediate

Banking and FinTech

CP-001 VaultCore: Core Banking Transaction Engine

Level: Foundation **Domain:** Banking and FinTech

A small cooperative bank still records deposits and withdrawals in paper ledgers, and the counter staff cannot tell a customer their balance without flipping pages. The bank wants a simple system for the counter.

Core Features (all 5 compulsory)

1. Open an account with an auto-generated account number, name, account type (savings or current) and opening deposit
2. Deposit and withdraw, rejecting withdrawals that would take the balance below the minimum balance for that account type
3. Show a mini statement of the last 5 transactions for an account
4. Search an account by account number or by name
5. Close an account only when its balance is zero, with a clear message otherwise

Stretch Features (optional)

- Credit monthly interest to all savings accounts at a rate the user enters
- Save all accounts and transactions to a file and load them when the program starts

Interview talking points: Validating money operations, keeping transaction history per account, enforcing business rules before changing state.

CP-002 SpendSense: Smart Budget Monitoring System with Threshold Alerts

Level: Foundation **Domain:** Banking and FinTech

Most students and young professionals run out of money before month-end without knowing where it went. A simple tracker that warns before a budget is crossed changes behaviour.

Core Features (all 5 compulsory)

1. Add an expense with date, category, amount and a short note
2. Set a monthly budget for each category
3. Warn when a category crosses 80 percent of its budget, and again at 100 percent
4. Show a month summary: total spent per category against budget
5. List the 5 largest expenses of the month

Stretch Features (optional)

- Compare this month with last month, category by category
- Export the month's expenses to a CSV file

Interview talking points: Aggregation by category, threshold alerts, sorting by amount, date handling.

CP-003 LoanLens: EMI Amortisation and Loan Comparison Engine

Level: Foundation **Domain:** Banking and FinTech

Borrowers compare loans only by interest rate and miss the total cost. A tool that shows the full repayment picture helps people choose the cheaper loan.

Core Features (all 5 compulsory)

1. Calculate the monthly EMI for a principal, annual interest rate and tenure in months
2. Print the full amortisation schedule: month, EMI, interest part, principal part, balance
3. Compare up to 3 loan offers and show which has the lowest total interest
4. Show the effect of a one-time prepayment on the remaining tenure
5. Reject invalid inputs such as negative principal, zero tenure or rates above 50 percent

Stretch Features (optional)

- Include processing fee and show the effective cost of each loan
- Save the amortisation schedule to a file

Interview talking points: Implementing a financial formula correctly, rounding of money values, tabular console output.

CP-004 FraudShield: Rule-Based UPI Transaction Risk Engine

Level: Intermediate **Domain:** Banking and FinTech

A payments startup wants simple, explainable fraud rules running before it invests in machine learning. Every flagged transaction must show which rule fired.

Core Features (all 5 compulsory)

1. Record transactions with id, user, amount, time (hh:mm) and city
2. Flag a transaction above 5 times that user's average amount
3. Flag a user making more than 3 transactions within 2 minutes
4. Flag a transaction from a city the user has never transacted in before
5. Print all flagged transactions with the reason, and a risk count per user

Stretch Features (optional)

- Let an admin change the rule thresholds without changing code
- Let an analyst mark a flag as a false alarm and report the false-alarm rate per rule

Interview talking points: Rule engines, time-window checks, per-user history, explainable decisions.

CP-005 CreditPulse: Explainable Credit Score Estimation Engine

Level: Intermediate **Domain:** Banking and FinTech

Young earners are rejected for loans without knowing why. A tool that estimates a credit score and explains what pulls it down helps them fix the right thing.

Core Features (all 5 compulsory)

1. Take inputs: on-time payment percentage, credit utilisation, age of oldest account, recent loan enquiries and number of credit types
2. Compute a weighted score between 300 and 900 using fixed, documented weights
3. Classify the score into Poor, Fair, Good or Excellent
4. Explain the two factors pulling the score down the most
5. Run a what-if: show the new score if utilisation drops to a value the user enters

Stretch Features (optional)

- Track the score month by month and show the trend
- Give a short improvement plan based on the weakest factors

Interview talking points: Weighted scoring models, explaining a result, what-if simulation.

CP-006 PortfolioIQ: Equity Portfolio and P&L Analytics System

Level: Intermediate **Domain:** Banking and FinTech

Retail investors buy the same stock several times at different prices and lose track of their real average price and profit.

Core Features (all 5 compulsory)

1. Record buy and sell trades with stock symbol, quantity, price and date
2. Show current holdings with average buy price per stock
3. Take today's prices and show profit or loss per stock and in total
4. Separate realised profit (from sales) and unrealised profit (on holdings)
5. Show allocation by sector as a percentage of the portfolio

Stretch Features (optional)

- Alert when any holding falls more than 10 percent below its average price
- Export a portfolio statement to a file

Interview talking points: Weighted averages, state that changes with every trade, realised vs unrealised accounting.

CP-007 EvenSplit: Group Expense Settlement Optimiser

Level: Intermediate **Domain:** Banking and FinTech

Friends on a trip pay for different things and end up with a messy chain of who owes whom. The app should settle everything in the fewest payments.

Core Features (all 5 compulsory)

1. Create a group and add members
2. Add a shared expense paid by one member and split equally among chosen members
3. Show each member's net balance (owes or is owed)
4. Generate a settlement plan with the minimum number of payments
5. Show the full expense history of the group

Stretch Features (optional)

- Support unequal splits by percentage or exact amounts
- Support multiple groups with separate balances

Interview talking points: Greedy settlement algorithm, net-balance calculation, keeping totals consistent.

CP-008 CashFlowX: ATM Cash Dispensing and Security Simulator

Level: Foundation **Domain:** Banking and FinTech

A bank wants to test its ATM logic before deployment: PIN security, daily limits and dispensing the exact amount with the notes available.

Core Features (all 5 compulsory)

1. Log in with card number and PIN, blocking the card after 3 wrong attempts
2. Withdraw an amount using the fewest notes from denominations 500, 200 and 100
3. Track the ATM's note inventory and refuse amounts it cannot dispense exactly
4. Enforce a daily withdrawal limit per card
5. Show balance enquiry and the last 3 withdrawals

Stretch Features (optional)

- Admin mode to refill notes and view inventory
- Print a transaction receipt to a file

Interview talking points: Greedy note selection and its edge cases, security attempts counter, limits.

CP-009 GoalVault: Goal-Based Savings and SIP Planning Engine

Level: Foundation **Domain:** Banking and FinTech

People save without a target and give up. A planner that converts a goal into a monthly saving amount keeps them on track.

Core Features (all 5 compulsory)

1. Create a goal with name, target amount and target date
2. Calculate the monthly saving needed, with and without an expected annual return
3. Record monthly contributions against a goal
4. Show progress percentage and amount remaining for every goal
5. Warn when a goal is behind schedule and show the new monthly amount needed

Stretch Features (optional)

- Adjust the target for inflation at a rate the user enters
- Show best, expected and worst return scenarios side by side

Interview talking points: Compound growth calculations, tracking progress against a plan, recalculation.

CP-010 TaxLedger: GST-Compliant Invoicing and Tax Engine

Level: Intermediate **Domain:** Banking and FinTech

Small shops in India must issue GST-compliant invoices. Many still write them by hand and make tax mistakes between CGST, SGST and IGST.

Core Features (all 5 compulsory)

1. Maintain a product list with code, name, price and GST rate (5, 12, 18 or 28 percent)
2. Create an invoice with multiple items and quantities
3. Apply CGST plus SGST for same-state buyers and IGST for other states
4. Generate invoice numbers in sequence and print a formatted invoice
5. Show the day's total sales and tax collected by rate

Stretch Features (optional)

- Issue a credit note against an earlier invoice
- Save each invoice as a text file

Interview talking points: Tax rules stored as data rather than hard-coded, formatted output, sequence generation.

E-Commerce and Retail

CP-011 SwiftCart: Point-of-Sale Billing and Inventory System

Level: Foundation **Domain:** E-Commerce and Retail

A neighbourhood supermarket has long queues because billing is manual. It needs a fast counter system that also keeps stock correct.

Core Features (all 5 compulsory)

1. Look up products by code and add them to the cart with quantity
2. Apply offers such as buy 2 get 1 free and a percentage discount on selected items
3. Print a bill with item lines, discounts, tax and grand total
4. Reduce stock for every sale and refuse items that are out of stock
5. Show end-of-day sales total and the number of bills

Stretch Features (optional)

- Loyalty points earned per bill and redeemable on the next bill
- Low-stock report for items below a reorder level

Interview talking points: Cart handling, applying offer rules, keeping inventory in sync with sales.

CP-012 StockSense: Predictive Inventory Reorder Engine

Level: Intermediate **Domain:** E-Commerce and Retail

A distributor either runs out of fast-moving items or overstocks slow ones. It needs to know what to reorder, when and how much.

Core Features (all 5 compulsory)

1. Maintain items with stock, average daily usage, supplier lead time in days and reorder level
2. Record stock-in and stock-out movements with date
3. Flag items whose stock will run out before a new order can arrive
4. Suggest an order quantity for each flagged item to cover a chosen number of days
5. Show the full movement history of any item

Stretch Features (optional)

- ABC classification of items by the value they consume
- Track batches with expiry dates and flag expired stock

Interview talking points: Reorder-point logic, forecasting from averages, inventory movement logs.

CP-013 BuyNext: Co-Purchase Product Recommendation Engine

Level: Advanced **Domain:** E-Commerce and Retail

An online store wants a "customers who bought this also bought" feature without any machine learning platform.

Core Features (all 5 compulsory)

1. Store orders as a customer plus the list of products bought together
2. Build co-purchase counts: how often each pair of products appears in the same order
3. For a given product, show the top 3 products most often bought with it
4. For a given customer, recommend 3 products they have not bought, based on what they did buy
5. Fall back to overall best sellers for a customer with no history

Stretch Features (optional)

- Find the most similar customer to a given customer using shared purchases
- Show recommendation reasons, for example "bought with X in 12 orders"

Interview talking points: Co-occurrence counting, 2D counting tables, recommendation basics and cold-start problem.

CP-014 DealForge: Coupon Validation and Best-Offer Optimisation Engine

Level: Intermediate **Domain:** E-Commerce and Retail

An e-commerce site runs many coupons with different conditions and customers get confused about why a coupon fails. The engine must apply the best valid coupon and explain rejections.

Core Features (all 5 compulsory)

1. Define coupons: flat or percentage off, minimum cart value, maximum discount, expiry date and category
2. Check a cart against one coupon and either apply it or explain exactly why it is not valid
3. Automatically find and apply the best valid coupon for a cart
4. Allow each coupon to be used only once per customer
5. Report how many times each coupon was used and the total discount given

Stretch Features (optional)

- Allow defined pairs of coupons to be stacked
- Coupons valid only on certain weekdays or time windows

Interview talking points: Validation rules with clear error messages, choosing the best option, usage tracking.

CP-015 RushQueue: Fair-Allocation Flash Sale Order Engine

Level: Advanced **Domain:** E-Commerce and Retail

During a flash sale, thousands of orders arrive for limited stock. The system must be fair, first come first served, and handle cancellations.

Core Features (all 5 compulsory)

1. Set up a sale with item, stock quantity and per-customer limit
2. Accept orders with customer and timestamp, confirming in order of arrival until stock ends
3. Put orders after sell-out on a waitlist
4. On a cancellation, release the stock to the next waitlisted customer automatically
5. Print the final report: confirmed, waitlisted, cancelled and rejected orders with reasons

Stretch Features (optional)

- Reject duplicate orders from the same customer within 10 seconds
- Simulate 500 random orders and report how long the stock lasted

Interview talking points: Queues and fairness, state transitions, handling contention for limited stock.

CP-016 ReturnFlow: Policy-Driven Returns and Refund Management System

Level: Foundation **Domain:** E-Commerce and Retail

An online retailer receives many return requests and processes them inconsistently. It needs one set of rules applied to every return.

Core Features (all 5 compulsory)

1. Look up an order by order id and show its items
2. Check return eligibility by days since delivery and product category rules
3. Record a return with a reason code from a fixed list
4. Calculate the refund after deducting shipping or restocking charges where applicable
5. Track return status and summarise returns by reason

Stretch Features (optional)

- Flag customers with more than 3 returns in 30 days
- Exchange instead of refund, checking stock of the replacement

Interview talking points: Policy rules as conditions, status tracking, reason analytics.

CP-017 TrackPoint: Order Lifecycle and Delivery Tracking System

Level: Foundation **Domain:** E-Commerce and Retail

Customer support spends most of its time answering "where is my order". A tracker with clear statuses reduces those calls.

Core Features (all 5 compulsory)

1. Place an order with customer, items and delivery pincode
2. Move an order through Placed, Packed, Shipped, Out for Delivery and Delivered, allowing only valid next steps
3. Track any order by id with its full status history
4. List orders that have been in one status for more than 2 days
5. Show a customer's complete order history

Stretch Features (optional)

- Estimate the delivery date from the pincode zone
- Allow cancellation only before the order is shipped

Interview talking points: Modelling a state machine, rejecting invalid transitions, history logs.

CP-018 PriceRadar: Multi-Seller Price Comparison and Alert Engine

Level: Intermediate **Domain:** E-Commerce and Retail

Shoppers check the same product on several sites to find the best deal. A tool that compares sellers, including delivery charges, saves time.

Core Features (all 5 compulsory)

1. Store products with prices from at least 3 sellers, plus delivery charge and delivery days
2. Show the cheapest total price (price plus delivery) for a product
3. Keep a price history and show the lowest price ever seen
4. Set a target price and alert when any seller drops to it
5. Search and sort products by name, price or delivery days

Stretch Features (optional)

- Rank sellers by a combined score of price and delivery time
- Show price trend over the last 5 updates as a text chart

Interview talking points: Comparison logic, history tracking, sorting by multiple keys.

CP-019 LoyaltyLoop: Tiered Customer Rewards Management System

Level: Foundation **Domain:** E-Commerce and Retail

A retail chain wants repeat customers and plans a points programme with Silver, Gold and Platinum tiers.

Core Features (all 5 compulsory)

1. Enrol a customer with phone number as the unique id
2. Earn points on every purchase, at a rate depending on the tier
3. Redeem points against a purchase, with a minimum redemption balance
4. Upgrade or downgrade tiers based on spend in the last 12 months
5. Show a customer's points statement

Stretch Features (optional)

- Expire points older than one year
- Report how many customers are in each tier and the points liability

Interview talking points: Tier rules, ledger of points, upgrade and downgrade logic.

CP-020 SalesScope: Retail Sales Analytics Dashboard

Level: Intermediate **Domain:** E-Commerce and Retail

A store owner gets a pile of daily sales records and no time to analyse them. He wants the key numbers in one screen.

Core Features (all 5 compulsory)

1. Load or enter sales records: date, product, category, quantity and amount
2. Show daily and weekly sales totals
3. Show the top 5 products by revenue and by quantity
4. Show each category's share of sales as a text bar chart
5. Compare two chosen weeks and show the change in percent

Stretch Features (optional)

- Identify the best and worst day of the week on average
- Export the dashboard summary to a text file

Interview talking points: Aggregation and grouping, ranking, presenting data clearly in text.

Healthcare

CP-021 MediQueue: Priority-Based Patient Queue Management System

Level: Foundation **Domain:** Healthcare

A busy clinic has patients crowding the reception asking when their turn is. Tokens with clear priority rules calm the waiting room.

Core Features (all 5 compulsory)

1. Register a patient and issue a token for a chosen doctor
2. Give priority to emergency cases and senior citizens without starving the normal queue
3. Call the next patient for each doctor
4. Show the estimated wait for any token based on average consultation time
5. Print a day summary: patients seen per doctor and average wait

Stretch Features (optional)

- Allow a patient to be skipped and recalled later
- Show live queue position for a token number

Interview talking points: Priority queues, fairness rules, wait-time estimation.

CP-022 PharmaTrack: Batch-Level Medicine Inventory and Expiry Control System

Level: Foundation **Domain:** Healthcare

Pharmacies lose money to expired medicines and risk selling them by mistake. Stock must be sold earliest-expiry first.

Core Features (all 5 compulsory)

1. Add medicine stock by batch with expiry date and quantity
2. Sell a quantity, always taking from the batch that expires first
3. Alert for batches expiring within 30 days
4. Show low-stock medicines below a reorder level
5. Block the sale of any expired batch

Stretch Features (optional)

- Flag prescription-only medicines and ask for a prescription number before sale
- Value the stock that expired this month

Interview talking points: First-expiry-first-out logic, date comparison, multi-batch stock.

CP-023 LifeLink: Blood Inventory and Donor Matching System

Level: Intermediate **Domain:** Healthcare

Hospitals lose time finding compatible blood in emergencies. A blood bank needs to match requests to stock and to eligible donors quickly.

Core Features (all 5 compulsory)

1. Register donors with blood group, phone and last donation date
2. Maintain stock of units by blood group with collection date
3. For a request, list compatible blood groups available in stock, exact match first
4. Find donors eligible to donate today (90 days since last donation)
5. Show a shortage report for groups below a minimum level

Stretch Features (optional)

- Expire units older than 35 days automatically
- Record which units were issued to which hospital

Interview talking points: Compatibility rules as a table, eligibility checks with dates, matching logic.

CP-024 VitalWatch: Patient Vitals Monitoring and Early-Alert System

Level: Intermediate **Domain:** Healthcare

Ward nurses record vitals on paper and early warning signs get missed during shift changes.

Core Features (all 5 compulsory)

1. Record vitals for a patient: blood pressure, pulse, temperature and oxygen saturation, with time
2. Raise an alert when any value is outside its normal range
3. Show the trend for a patient over the last 3 readings (improving, stable or worsening)
4. List all patients with an active alert
5. Print a shift handover summary for the ward

Stretch Features (optional)

- Calculate a simple early-warning score combining all vitals
- Escalate a patient whose score rises in two consecutive readings

Interview talking points: Threshold checks, trend detection, summarising for a handover.

CP-025 BedFlow: Hospital Bed Allocation and Occupancy Management System

Level: Intermediate **Domain:** Healthcare

During peak days, hospitals struggle to know which beds are free and patients wait in corridors.

Core Features (all 5 compulsory)

1. Set up wards with bed types: general, semi-private and ICU
2. Admit a patient to a free bed of the required type
3. Discharge a patient and free the bed
4. Show occupancy percentage per ward and overall
5. Keep a waitlist when no bed of the required type is free, and allocate in order when one frees up

Stretch Features (optional)

- Transfer a patient between wards
- Calculate the bill by bed type and number of days

Interview talking points: Resource allocation, waitlists, occupancy reporting.

CP-026 NutriPlan: Personalised Calorie and Macro Planning Engine

Level: Foundation **Domain:** Healthcare

Fitness apps are popular with students, but most people never understand the numbers behind them. This planner makes the calculation visible.

Core Features (all 5 compulsory)

1. Maintain a food list with calories, protein, carbohydrates and fat per serving
2. Log meals for a day with servings
3. Calculate the daily calorie target from age, height, weight, gender and activity level using a standard formula
4. Show the day's totals against the target
5. Show a weekly summary with average intake

Stretch Features (optional)

- Suggest foods to close the protein gap for the day
- Track weight entries and show the trend

Interview talking points: Applying a published formula, aggregating nutrients, daily and weekly summaries.

CP-027 DoseMate: Medication Scheduling and Adherence Tracker

Level: Foundation **Domain:** Healthcare

Elderly patients and caregivers struggle to track multiple medicines at different times of day.

Core Features (all 5 compulsory)

1. Add a medicine with dosage, times of day and number of days
2. Generate today's schedule in time order
3. Mark each dose as taken or missed
4. Show the adherence percentage per medicine
5. Alert when the remaining tablets will last fewer than 3 days

Stretch Features (optional)

- Support medicines taken on alternate days or weekly
- Weekly adherence report for a caregiver

Interview talking points: Schedule generation, time ordering, adherence tracking.

CP-028 LabInsight: Diagnostic Report Analysis and Trend Engine

Level: Intermediate **Domain:** Healthcare

Patients receive lab reports full of numbers and do not know what is normal. The analyzer highlights what needs attention.

Core Features (all 5 compulsory)

1. Maintain reference ranges for common tests, by gender where needed
2. Enter a patient's results for a report
3. Flag each value as low, normal or high
4. Print a clean report with flagged values clearly marked
5. Compare with the patient's previous report and show what changed

Stretch Features (optional)

- Group tests into panels such as lipid profile or blood count
- Save each patient's reports to a file

Interview talking points: Range lookups, comparison across records, readable report formatting.

CP-029 ClaimSure: Health Insurance Claim Adjudication Engine

Level: Intermediate **Domain:** Healthcare

Insurance claims are delayed and customers do not understand partial approvals. The tracker applies policy rules and explains every deduction.

Core Features (all 5 compulsory)

1. Store policies with sum insured, room-rent limit, waiting period and amount already claimed this year
2. Submit a claim with treatment, hospital bill and dates
3. Check waiting period and remaining sum insured
4. Approve fully, approve partially with each deduction explained, or reject with a reason
5. Track claim status and show a policyholder's claim history

Stretch Features (optional)

- Apply co-payment rules for senior citizens
- Report total claims paid by treatment type

Interview talking points: Applying several rules in sequence, explaining calculations, status tracking.

CP-030 OutbreakSim: Grid-Based Epidemic Spread Simulation Model

Level: Advanced **Domain:** Healthcare

Public health teams use simulations to show how vaccination and distancing change the spread of a disease. This is a simple grid-based version.

Core Features (all 5 compulsory)

1. Model a population as a grid where each cell is susceptible, infected, recovered or vaccinated
2. Each day, infected cells infect neighbours with a given probability
3. Infected cells recover after a set number of days
4. Run for a given number of days and print daily counts of each state
5. Compare two scenarios, for example 0 percent and 50 percent vaccination

Stretch Features (optional)

- Print the grid as text for chosen days
- Find the peak day and peak number of infections

Interview talking points: 2D arrays, simulation loops, randomness, comparing scenarios.

Education and Campus

CP-031 GradeCore: Academic Result Processing and Ranking Engine

Level: Foundation **Domain:** Education and Campus

Departments still calculate results in spreadsheets with manual errors. A result system needs to be accurate and fast.

Core Features (all 5 compulsory)

1. Enter marks for students across 5 subjects with validation (0 to 100)
2. Calculate total, percentage and grade for every student
3. Mark pass or fail per subject using the minimum pass mark
4. Rank students by percentage, handling ties properly
5. Print a report card for any student and the topper for each subject

Stretch Features (optional)

- Calculate SGPA using credits per subject
- Show the class distribution by grade as a text chart

Interview talking points: Validation, ranking with ties, report formatting.

CP-032 AttendIQ: Attendance Analytics and Shortage Prediction System

Level: Foundation **Domain:** Education and Campus

Students discover attendance shortage only when hall tickets are withheld. An early warning helps them act in time.

Core Features (all 5 compulsory)

1. Record daily attendance per subject for a class
2. Calculate attendance percentage per student per subject
3. List students below 75 percent in any subject
4. Calculate how many consecutive classes a student must attend to reach 75 percent
5. Print a weekly attendance report for the class

Stretch Features (optional)

- Mark medical leave days that do not count against the student
- Predict which students will fall below 75 percent at the current rate

Interview talking points: Percentage calculations, simple projection, filtering by condition.

CP-033 QuizForge: Randomised Assessment and Leaderboard Engine

Level: Intermediate **Domain:** Education and Campus

Colleges want quick practice tests for placement topics without depending on paid platforms.

Core Features (all 5 compulsory)

1. Maintain a question bank with options, correct answer, topic and difficulty
2. Generate a random quiz of N questions for a topic with no repeats
3. Score with negative marking for wrong answers
4. Show a review of every question with the student's answer and the correct one
5. Keep a leaderboard of top scores per topic

Stretch Features (optional)

- Adaptive mode: next question harder after a correct answer, easier after a wrong one
- Topic-wise performance report for a student

Interview talking points: Random selection without repetition, scoring rules, leaderboards.

CP-034 ClashGuard: Timetable Conflict Detection and Resolution Engine

Level: Advanced **Domain:** Education and Campus

Every semester, timetables are built by hand and clashes are found only when classes start.

Core Features (all 5 compulsory)

1. Store courses with faculty, room, student group and weekly slots
2. Detect a faculty member booked in two places at the same time
3. Detect a room double-booked
4. Detect a student group with two classes in the same slot
5. Suggest free slots where a course can move without any clash

Stretch Features (optional)

- Print the weekly timetable grid for a group, a faculty member or a room
- Check faculty workload against a maximum hours per week

Interview talking points: Conflict detection, 2D grids for days and slots, searching for valid options.

CP-035 ShelfWise: Library Circulation and Reservation Management System

Level: Foundation **Domain:** Education and Campus

The college library tracks issues in a register, cannot find who has a book, and fines are calculated by hand.

Core Features (all 5 compulsory)

1. Add books with id, title, author and number of copies
2. Issue a book to a member if a copy is available and the member is under their limit
3. Return a book and calculate the fine for late days
4. Search books by title or author
5. List overdue books with member details

Stretch Features (optional)

- Reserve a book that is out, and notify the first reserver on return
- Report the most borrowed books of the month

Interview talking points: Issue and return lifecycle, limits, date-based fines, search.

CP-036 SeatPlanner: Constraint-Based Exam Seating Allocation Engine

Level: Intermediate **Domain:** Education and Campus

For internal exams, the exam cell arranges seating so that students of the same branch do not sit next to each other.

Core Features (all 5 compulsory)

1. Store rooms with rows and columns of seats
2. Take student lists from multiple branches
3. Allocate seats so no two adjacent students (left or right) are from the same branch
4. Print the seat map for each room
5. Look up a student's room and seat by register number

Stretch Features (optional)

- Also avoid same-branch students directly in front or behind
- Generate a door sheet listing register numbers per room

Interview talking points: 2D seat arrays, constraint-based allocation, lookup.

CP-037 RoomRight: Priority-Based Hostel Allocation System

Level: Intermediate **Domain:** Education and Campus

Hostel allocation is a yearly headache of preferences, priorities and complaints of unfairness.

Core Features (all 5 compulsory)

1. Store rooms with type (single, double, triple) and capacity
2. Accept applications with year, distance from home and room preference
3. Allocate by a clear priority order, giving preference where possible
4. Process swap requests between two students
5. Show vacancies and students still waiting

Stretch Features (optional)

- Track hostel fee dues per student
- Fairness report: how many students got their first preference

Interview talking points: Priority-based allocation, explaining fairness, swaps that keep data consistent.

CP-038 PlaceMatch: Multi-Criteria Placement Eligibility Engine

Level: Intermediate **Domain:** Education and Campus

Placement cells manually filter hundreds of students against each company's criteria, and eligible students get missed.

Core Features (all 5 compulsory)

1. Store student profiles: CGPA, active backlogs, branch, 10th and 12th percentages and skills
2. Store company criteria for each drive
3. List eligible students for a company
4. Show a student every company they are eligible for
5. Show drive statistics: eligible, applied and selected counts

Stretch Features (optional)

- Rank eligible students by how many required skills they have
- Explain to a student exactly which criterion they missed for a company

Interview talking points: Multi-criteria filtering, two-way lookup, explaining rejections.

CP-039 FeedbackIQ: Course Feedback Sentiment and Ratings Analyser

Level: Intermediate **Domain:** Education and Campus

Colleges collect feedback every semester but nobody reads hundreds of comments. The analyzer surfaces what matters.

Core Features (all 5 compulsory)

1. Record feedback: course, faculty, ratings on 5 parameters and a text comment
2. Average each parameter per course and faculty
3. Classify comments as positive, negative or neutral using keyword lists
4. Flag courses with any parameter average below 3
5. Print a report per faculty with strengths and areas to improve

Stretch Features (optional)

- Show the most frequent words in negative comments
- Compare two semesters for the same course

Interview talking points: Averages by group, simple text analysis with keywords, actionable reports.

CP-040 RecallPro: Spaced-Repetition Learning Scheduler

Level: Advanced **Domain:** Education and Campus

Spaced repetition is how language apps make learners remember. This trainer uses the same idea for placement and exam revision.

Core Features (all 5 compulsory)

1. Create decks of question and answer cards
2. Run a review session where the user rates each card easy, medium or hard
3. Schedule each card's next review date using a box system that spaces easy cards further apart
4. Show cards due today
5. Show stats: cards learnt, cards due and accuracy

Stretch Features (optional)

- Reset a card to the first box after it is forgotten twice
- Show a 7-day forecast of cards due

Interview talking points: Scheduling algorithms, date arithmetic, a known learning technique implemented from scratch.

Travel, Transport and Logistics

CP-041 RouteSeat: Intercity Bus Reservation and Seat Management System

Level: Foundation **Domain:** Travel, Transport and Logistics

A regional bus operator sells tickets at the counter and cannot see seat availability across routes.

Core Features (all 5 compulsory)

1. Set up routes and buses with seat layout
2. Show available seats as a seat map
3. Book seats with passenger name and age, fare based on distance
4. Cancel a booking with a refund depending on how early it is cancelled
5. Print the passenger list for a bus

Stretch Features (optional)

- Reserve certain seats for women and senior citizens
- Revenue report per route

Interview talking points: Seat maps in arrays, booking and cancellation rules, refunds.

CP-042 MetroMind: Shortest-Route and Fare Computation Engine

Level: Advanced **Domain:** Travel, Transport and Logistics

A metro system with two or more lines needs a tool that tells passengers the route with the fewest stops and the fare.

Core Features (all 5 compulsory)

1. Store stations and the lines they belong to, with interchange stations
2. Find the route with the fewest stops between two stations
3. Show the number of stops and the interchanges on the route
4. Calculate the fare by number of stations
5. Deduct the fare from a travel card balance and refuse travel when the balance is too low

Stretch Features (optional)

- Apply a 10 percent discount for travel in off-peak hours
- Show all stations within N stops of a station

Interview talking points: Modelling a network as a graph, shortest path by breadth-first search, fare rules.

CP-043 ParkSmart: Multi-Level Parking Allocation and Billing System

Level: Foundation **Domain:** Travel, Transport and Logistics

A mall parking lot has queues at entry and disputes over fees at exit.

Core Features (all 5 compulsory)

1. Set up levels with slots for two-wheelers, cars and EVs
2. Assign the nearest free slot of the right type on entry and issue a ticket
3. Calculate the fee on exit by duration with hourly slabs
4. Show live occupancy per level and type
5. Show the day's revenue and number of vehicles

Stretch Features (optional)

- Charge EV slots extra for charging time
- Handle a lost ticket with a fixed penalty

Interview talking points: Slot allocation, time-based fees, occupancy tracking.

CP-044 RideFare: Dynamic Surge Pricing Engine for Ride-Hailing

Level: Intermediate **Domain:** Travel, Transport and Logistics

Ride-hailing apps calculate fares from distance, time and demand. This estimator reproduces that logic transparently.

Core Features (all 5 compulsory)

1. Store vehicle types with base fare, per-km rate and per-minute rate
2. Estimate a fare from distance and expected time
3. Apply a surge multiplier from the ratio of ride requests to available drivers
4. Apply night charges between 11 PM and 5 AM
5. Show a fare breakdown and compare all vehicle types for the same trip

Stretch Features (optional)

- Keep a trip history and show monthly spend
- Cap the surge at a maximum multiplier

Interview talking points: Pricing rules, multipliers, transparent breakdowns.

CP-045 ShipTrail: Zone-Based Courier Tariff and Shipment Tracking System

Level: Intermediate **Domain:** Travel, Transport and Logistics

A courier company needs tariffs by zone and weight and a clear tracking trail for every parcel.

Core Features (all 5 compulsory)

1. Book a shipment with sender, receiver, pincodes and weight
2. Calculate the charge by weight slab and zone (local, regional, national)
3. Record scan events (picked, in transit, at hub, out for delivery, delivered) with place and time
4. Track a shipment by number with the full scan history
5. List shipments past their expected delivery date

Stretch Features (optional)

- Record proof of delivery with receiver name
- Cash-on-delivery amounts collected per day

Interview talking points: Slab pricing, event logs, delay detection.

CP-046 SkyBook: Dynamic-Fare Flight Reservation and Waitlist Engine

Level: Intermediate **Domain:** Travel, Transport and Logistics

Airlines sell seats at changing prices and manage waitlists when flights are full.

Core Features (all 5 compulsory)

1. Set up flights with economy and business seats
2. Book with a PNR number; fares rise in steps as seats fill
3. Put bookings on a waitlist when a class is full
4. On cancellation, confirm the first waitlisted passenger automatically
5. Look up a booking by PNR

Stretch Features (optional)

- Allow an upgrade from economy to business when seats are available
- Report load factor per flight

Interview talking points: Dynamic pricing, waitlist promotion, unique id generation.

CP-047 FleetPulse: Fleet Fuel Efficiency and Maintenance Analytics System

Level: Foundation **Domain:** Travel, Transport and Logistics

A small transport company does not know which vehicles waste fuel or are overdue for service.

Core Features (all 5 compulsory)

1. Register vehicles with number, type and current odometer reading
2. Record fuel fills with litres, cost and odometer reading
3. Calculate mileage per fill and average mileage per vehicle
4. Flag vehicles due for service by kilometres or by date
5. Show cost per km for each vehicle and the worst performer

Stretch Features (optional)

- Record maintenance costs and show total cost of ownership
- Flag a sudden drop in mileage

Interview talking points: Derived metrics, due-date logic, comparing records.

CP-048 LoadIQ: Priority-Driven Delivery Load Optimisation Engine

Level: Advanced **Domain:** Travel, Transport and Logistics

A logistics hub loads vans by hand and often leaves high-priority parcels behind while vans leave half-full.

Core Features (all 5 compulsory)

1. Record parcels with weight, volume and priority
2. Record vans with weight and volume capacity
3. Assign parcels to vans, highest priority first, without exceeding capacity
4. List parcels that could not be loaded and why
5. Print a load sheet per van with total weight and volume used

Stretch Features (optional)

- Compare a priority-first plan with a value-per-weight plan
- Balance load across vans as evenly as possible

Interview talking points: Greedy heuristics, capacity constraints, knapsack-style reasoning.

CP-049 PlatformPro: Interval-Scheduling Platform Allocation Engine

Level: Advanced **Domain:** Travel, Transport and Logistics

A station must assign platforms to trains so no two trains are on the same platform at the same time.

Core Features (all 5 compulsory)

1. Record trains with arrival and departure times
2. Calculate the minimum number of platforms needed for the day
3. Assign a platform to each train
4. Handle a delay by moving a train to another free platform if needed
5. Print the platform chart in time order

Stretch Features (optional)

- Keep certain platforms for long-distance trains only
- Report platform utilisation

Interview talking points: Interval scheduling, sorting by time, re-allocation after a change.

CP-050 TripCraft: Itinerary Scheduling and Budget Control Planner

Level: Foundation **Domain:** Travel, Transport and Logistics

Group trips go over budget and schedules clash. A planner keeps both in check.

Core Features (all 5 compulsory)

1. Create a trip with dates, members and total budget
2. Add activities per day with start time, end time and cost
3. Detect overlapping activities on the same day
4. Compare planned cost with actual spend as it happens
5. Show the day-wise itinerary and per-person cost

Stretch Features (optional)

- Save the itinerary to a text file to share
- Suggest which activities to drop to get back under budget

Interview talking points: Time overlap checks, budget tracking, clear schedules.

HR and Workplace

CP-051 LeaveFlow: Leave Approval Workflow and Team Availability System

Level: Foundation **Domain:** HR and Workplace

Leave requests go by email and WhatsApp, balances are wrong, and two people from the same team go on leave together.

Core Features (all 5 compulsory)

1. Maintain leave types and balances per employee (casual, sick, earned)
2. Apply for leave with dates and type, rejecting requests beyond the balance
3. Approve or reject requests as a manager, updating balances only on approval
4. Warn when more than 2 members of the same team are on leave on the same day
5. Show a month calendar of who is on leave

Stretch Features (optional)

- Carry forward unused earned leave at year end up to a limit
- Exclude weekends and public holidays from leave days

Interview talking points: Approval workflows, balance consistency, overlap checks on dates.

CP-052 TalentTrack: Skill-Matching Applicant Tracking System

Level: Intermediate **Domain:** HR and Workplace

A recruiter receives hundreds of resumes for every opening and needs to shortlist quickly and track candidates through the hiring stages.

Core Features (all 5 compulsory)

1. Create job openings with required skills and minimum experience
2. Add candidates with skills, experience and contact details
3. Score each candidate against a job by matching skills, and list them by score
4. Move candidates through stages: Applied, Screened, Interview, Offer, Joined or Rejected
5. Show the hiring funnel for a job: number of candidates at each stage

Stretch Features (optional)

- Treat related skills as matches, for example Java and Spring
- Show average days spent in each stage

Interview talking points: Matching and scoring, pipelines, funnel reporting. A good project to discuss with HR interviewers.

CP-053 ShiftGenie: Constraint-Based Workforce Roster Generator

Level: Advanced **Domain:** HR and Workplace

A hospital or support centre builds weekly rosters by hand, causing unfair shift loads and understaffed nights.

Core Features (all 5 compulsory)

1. Record staff with availability and maximum hours per week
2. Define shifts with the number of people needed
3. Generate a weekly roster that fills shifts while respecting availability and maximum hours
4. Distribute night shifts as evenly as possible
5. List any shift that could not be fully staffed

Stretch Features (optional)

- Process shift swap requests between two staff members, checking rules
- Enforce a minimum rest gap between two shifts

Interview talking points: Constraint-based scheduling, fairness heuristics, reporting gaps.

CP-054 PunchLog: Rule-Based Attendance and Overtime Engine

Level: Foundation **Domain:** HR and Workplace

An office tracks check-in times but applies late-coming rules inconsistently, which causes disputes at salary time.

Core Features (all 5 compulsory)

1. Record daily check-in and check-out times for employees
2. Mark late arrivals after a grace time
3. Convert every 3 late marks in a month into a half-day deduction
4. Calculate overtime hours beyond the standard day
5. Print a monthly attendance summary per employee

Stretch Features (optional)

- Support flexible shift timings per employee
- Flag missing check-outs

Interview talking points: Time calculations, monthly rule application, summary reports.

CP-055 RoomSync: Conflict-Free Meeting Room Booking System

Level: Foundation **Domain:** HR and Workplace

Teams fight over meeting rooms and some rooms sit empty because nobody knows they are free.

Core Features (all 5 compulsory)

1. Store rooms with capacity and equipment (projector, video call)
2. Book a room for a time slot without overlapping existing bookings
3. Find a free room for a given time, number of people and equipment
4. Cancel a booking
5. Show room utilisation for the week

Stretch Features (optional)

- Support weekly recurring bookings
- Release a booking automatically if nobody checks in within 10 minutes

Interview talking points: Interval overlap checks, search with several conditions, utilisation.

CP-056 SkillGrid: Skill-Gap Analysis and Training Recommendation Engine

Level: Intermediate **Domain:** HR and Workplace

A services company cannot see which employees have which skills, so projects get staffed badly and training is random.

Core Features (all 5 compulsory)

1. Record employees with skills rated 1 to 5
2. Record project requirements as skills with minimum ratings
3. Find the best-fit employees for a project
4. Show the skill gap for a team against a project
5. Suggest who should be trained in what to close the gap

Stretch Features (optional)

- Track training completion and update ratings
- Show the company's strongest and weakest skills

Interview talking points: Matrix data, matching and gap analysis, recommendations.

CP-057 ClaimFlow: Multi-Level Expense Approval Workflow Engine

Level: Intermediate **Domain:** HR and Workplace

Employees wait weeks for reimbursement because claims bounce between managers and finance without clear rules.

Core Features (all 5 compulsory)

1. Submit a claim with category, amount, date and whether a receipt is attached
2. Check category limits and flag claims above the limit
3. Route claims for approval: manager only below a set amount, manager then finance above it
4. Track status at every step
5. Report policy violations and total reimbursement per employee

Stretch Features (optional)

- Reject duplicate claims for the same date, category and amount
- Monthly spend report by category

Interview talking points: Multi-level workflows, rule checks, audit trail.

CP-058 AssetTrail: IT Asset Lifecycle and Allocation Tracker

Level: Foundation **Domain:** HR and Workplace

A company loses track of laptops and monitors when employees move teams or leave.

Core Features (all 5 compulsory)

1. Register assets with serial number, type, purchase date and warranty end date
2. Assign an asset to an employee and record the return
3. Show the full history of any asset
4. List assets whose warranty ends within 60 days
5. List unassigned assets by type

Stretch Features (optional)

- Block an employee exit until all their assets are returned
- Show asset value after yearly depreciation

Interview talking points: Asset lifecycle tracking, history logs, date-based alerts.

CP-059 GatePass: Visitor Access and Security Management System

Level: Foundation **Domain:** HR and Workplace

Office security keeps a paper register, so nobody knows who is inside the building at any time.

Core Features (all 5 compulsory)

1. Record a visitor's entry with name, phone, host employee and purpose
2. Issue a visitor pass number
3. Record exit time
4. Show everyone currently inside
5. Check every new visitor against a blocked list

Stretch Features (optional)

- Pre-registration of expected visitors by employees
- Report frequent visitors and average visit duration

Interview talking points: Entry and exit tracking, live lists, validation against a list.

CP-060 ResolveDesk: SLA-Driven Helpdesk Ticketing System

Level: Intermediate **Domain:** HR and Workplace

IT and HR requests come through chats and get lost. A ticketing system with service-level timelines gives accountability.

Core Features (all 5 compulsory)

1. Raise a ticket with category, priority and description
2. Assign tickets to agents of that category in round-robin order
3. Set a resolution deadline by priority
4. Update status: Open, In Progress, Resolved or Closed
5. Report tickets that breached their deadline, per agent

Stretch Features (optional)

- Escalate automatically to a lead when a deadline is breached
- Collect a 1 to 5 rating when a ticket is closed

Interview talking points: Round-robin assignment, deadline tracking, accountability reports.

Food, Hospitality and Events

CP-061 KitchenSync: Restaurant Order Routing and Kitchen Display System

Level: Intermediate **Domain:** Food, Hospitality and Events

In a busy restaurant, waiters shout orders to the kitchen and items get missed or delayed.

Core Features (all 5 compulsory)

1. Maintain the menu with price and preparation time
2. Take orders per table with items and quantities
3. Show the kitchen queue in order time with each item's status
4. Mark items ready and show waiting tables
5. Generate a table bill with tax and a split into equal parts

Stretch Features (optional)

- Flag orders waiting longer than their preparation time plus 10 minutes
- Show the day's best-selling items

Interview talking points: Order queues, status tracking, billing.

CP-062 StayEase: Hotel Reservation and Availability Engine

Level: Foundation **Domain:** Food, Hospitality and Events

A small hotel takes bookings over the phone and has been double-booking rooms.

Core Features (all 5 compulsory)

1. Store rooms with type and nightly rate
2. Check availability for a date range
3. Book a room without overlapping existing bookings
4. Check out a guest and generate a bill with room charges and extras
5. Show occupancy for any date

Stretch Features (optional)

- Weekend and season pricing
- Cancellation with a refund policy

Interview talking points: Date-range overlap, availability search, billing.

CP-063 DashDispatch: Nearest-Rider Delivery Assignment Engine

Level: Advanced **Domain:** Food, Hospitality and Events

A food delivery platform must assign each new order to a rider quickly so food arrives hot.

Core Features (all 5 compulsory)

1. Record restaurants, customers and riders with positions on a city grid (x, y)
2. Assign each new order to the nearest free rider
3. Estimate delivery time from distances and an average speed
4. Free the rider when the order is delivered
5. Show each rider's deliveries and earnings for the day

Stretch Features (optional)

- Batch two orders from the same restaurant going in a similar direction
- Report average delivery time per area

Interview talking points: Nearest-neighbour assignment, distance calculation, real-time state.

CP-064 TicketVerse: Seat-Category Event Ticketing System

Level: Foundation **Domain:** Food, Hospitality and Events

A college fest or concert needs to sell seats by category and stop overselling.

Core Features (all 5 compulsory)

1. Set up a venue with sections, rows and price per section
2. Show available seats as a seat map
3. Book selected seats and generate a ticket id
4. Cancel tickets with a refund depending on days left to the event
5. Show sales by section and total revenue

Stretch Features (optional)

- Early-bird pricing until a date
- Group booking discount above 5 seats

Interview talking points: Seat maps, booking rules, refunds, revenue reporting.

CP-065 CookScale: Recipe Scaling and Smart Grocery Planner

Level: Foundation **Domain:** Food, Hospitality and Events

Home cooks and canteens plan meals for different numbers of people and always buy the wrong quantities.

Core Features (all 5 compulsory)

1. Store recipes with ingredients, quantities, units and servings
2. Scale a recipe to any number of servings
3. Plan meals for a week by choosing recipes
4. Build one combined grocery list, adding up the same ingredient across recipes
5. Estimate the grocery cost from a price list

Stretch Features (optional)

- Convert units, for example grams to kilograms and teaspoons to millilitres
- Remove items already in the pantry from the list

Interview talking points: Scaling, merging and summing records, unit handling.

CP-066 QuickBite: Slot-Based Canteen Pre-Ordering System

Level: Foundation **Domain:** Food, Hospitality and Events

The college canteen is overcrowded during breaks. Pre-ordering with pickup slots spreads the rush.

Core Features (all 5 compulsory)

1. Maintain the menu with price and preparation time
2. Place a pre-order with a pickup slot
3. Limit the number of orders per slot
4. Issue a token and show orders due in each slot for the kitchen
5. Show daily sales and the busiest slot

Stretch Features (optional)

- Forecast tomorrow's demand per item from the last 5 days' average
- Wallet balance per student, deducted on ordering

Interview talking points: Capacity-limited slots, simple forecasting, kitchen planning.

CP-067 QuoteChef: Catering Cost Estimation and Quotation Engine

Level: Foundation **Domain:** Food, Hospitality and Events

A catering business spends hours preparing quotes and makes costing mistakes that eat into margins.

Core Features (all 5 compulsory)

1. Store menu packages with per-plate cost
2. Take event type, number of guests and package
3. Calculate staff needed using a guests-per-staff ratio, and their cost
4. Add tax and margin to produce the final quote
5. Print a formatted quote and compare packages side by side

Stretch Features (optional)

- Discount slabs for large events
- Save quotes and track which were accepted

Interview talking points: Costing logic, ratios, formatted documents.

CP-068 GuestGrid: Event Guest and Table Seating Manager

Level: Foundation **Domain:** Food, Hospitality and Events

Families planning weddings or large events juggle guest lists, RSVPs and seating on paper.

Core Features (all 5 compulsory)

1. Maintain guests with group (family or friends), RSVP status and meal preference
2. Update RSVPs and show confirmed headcount
3. Allocate confirmed guests to tables of fixed capacity, keeping a group together where possible
4. Show the headcount by meal preference for the caterer
5. Print the seating chart

Stretch Features (optional)

- Mark guests who must not be seated at the same table
- Track gifts received and thank-you notes sent

Interview talking points: Grouping with capacity constraints, counts by category.

CP-069 TableTurn: Best-Fit Table Reservation and Waitlist Engine

Level: Intermediate **Domain:** Food, Hospitality and Events

A popular restaurant turns away walk-ins while reserved tables sit empty because of no-shows.

Core Features (all 5 compulsory)

1. Store tables with seating capacity
2. Book reservations for a time and party size, choosing the smallest table that fits
3. Keep a walk-in waitlist with an estimated waiting time
4. Release a reservation as a no-show after 15 minutes and seat the next waitlisted party
5. Show table turnover for the day

Stretch Features (optional)

- Join two tables for a large party
- Send waitlisted parties a turn number they can check

Interview talking points: Best-fit allocation, waitlists, time-based rules.

CP-070 EventLedger: Event Budget and Vendor Payment Tracker

Level: Intermediate **Domain:** Food, Hospitality and Events

Event organisers lose track of quotes, advances and balances across many vendors and overshoot the budget.

Core Features (all 5 compulsory)

1. Set budget heads such as venue, food, decoration and sound
2. Record vendor quotes per head and select one
3. Record payments in milestones (advance, balance)
4. Show budget against committed and paid amounts per head
5. Alert when a head goes over budget

Stretch Features (optional)

- List pending payments by due date
- Compare vendors on price and rating

Interview talking points: Budget tracking, payment milestones, variance alerts.

Public Services, Utilities and Smart City

CP-071 PowerBill: Slab-Based Utility Billing Engine

Level: Foundation **Domain:** Public Services, Utilities and Smart City

Consumers often do not understand how slab tariffs make their bill jump. The calculator shows exactly how the bill is built.

Core Features (all 5 compulsory)

1. Store consumer types (domestic, commercial) with slab rates and fixed charges
2. Take previous and current meter readings and calculate units
3. Calculate the bill slab by slab and show the breakdown
4. Add a late payment fee after the due date
5. Compare consumption with the same month last year

Stretch Features (optional)

- Apply a subsidy for the first slab for eligible consumers
- Show which consumers had a spike of more than 50 percent

Interview talking points: Slab calculations, transparent breakdowns, comparison with history.

CP-072 AquaRoute: Priority-Based Water Supply Scheduling System

Level: Intermediate **Domain:** Public Services, Utilities and Smart City

In water-scarce areas, residents request tankers and supply is allocated unevenly.

Core Features (all 5 compulsory)

1. Record requests with area, litres needed and priority (hospital, school, residential)
2. Record tankers with capacity
3. Schedule a day's trips so higher priority requests are served first
4. Carry unserved requests to the next day
5. Report water supplied per area

Stretch Features (optional)

- Group nearby requests into one trip when capacity allows
- Show the average days a request waits, per area

Interview talking points: Priority scheduling, capacity planning, carry-over.

CP-073 CivicConnect: Complaint Routing and Escalation System

Level: Intermediate **Domain:** Public Services, Utilities and Smart City

Citizen complaints about potholes, garbage and street lights are lost between departments.

Core Features (all 5 compulsory)

1. Register a complaint with category, ward and description
2. Route it automatically to the right department by category
3. Update status: Received, Assigned, In Progress or Resolved
4. Escalate complaints pending longer than a set number of days
5. Show statistics per ward and per department

Stretch Features (optional)

- Detect duplicate complaints about the same issue in the same ward
- Rank departments by average resolution time

Interview talking points: Routing rules, escalation logic, public-service reporting.

CP-074 SignalSense: Adaptive Traffic Signal Simulation Model

Level: Advanced **Domain:** Public Services, Utilities and Smart City

Fixed signal timings cause long queues on busy roads while empty roads get green. Simulation shows whether adaptive timing helps.

Core Features (all 5 compulsory)

1. Model a junction with 4 approaches and vehicle arrival rates per approach
2. Simulate fixed green times cycle by cycle
3. Simulate adaptive green times proportional to the queue on each approach
4. Track queue length and average waiting time per approach
5. Compare fixed and adaptive results in a report

Stretch Features (optional)

- Add a pedestrian phase
- Simulate peak and off-peak arrival rates

Interview talking points: Simulation loops, queues, comparing two strategies with numbers.

CP-075 BinSense: Smart Waste Monitoring and Collection Route Planner

Level: Intermediate **Domain:** Public Services, Utilities and Smart City

Garbage trucks visit every bin on a fixed route, whether full or empty, wasting fuel while some bins overflow.

Core Features (all 5 compulsory)

1. Record bins with location (x, y) and fill level
2. Update fill levels
3. List bins above 75 percent full
4. Build a collection route visiting only those bins, each time going to the nearest unvisited bin
5. Alert on overflowing bins and keep a history of collections

Stretch Features (optional)

- Predict when a bin will be full from its fill rate
- Calculate the route distance saved against visiting every bin

Interview talking points: Nearest-neighbour routing, threshold filtering, prediction from rates.

CP-076 ChallanTrack: Traffic Violation and Penalty Management System

Level: Foundation **Domain:** Public Services, Utilities and Smart City

Traffic police need to issue fines quickly and identify repeat offenders.

Core Features (all 5 compulsory)

1. Maintain offences with fine amounts
2. Issue a challan to a vehicle number after checking the number format
3. Increase the fine for repeat offences by the same vehicle
4. Record payment of a challan
5. List pending challans by vehicle and total amount due

Stretch Features (optional)

- Suspend a licence after a set number of offences in a year
- Report the most common offences by area

Interview talking points: Format validation, repeat-offender logic, payment tracking.

CP-077 RationRight: Entitlement-Based Public Distribution Tracker

Level: Intermediate **Domain:** Public Services, Utilities and Smart City

Fair price shops must give each family its entitlement exactly once a month and keep stock accounts accurate.

Core Features (all 5 compulsory)

1. Register families with card type and members
2. Calculate monthly entitlement of rice, wheat and sugar by card type and members
3. Record distribution and block a second collection in the same month
4. Track shop stock and refuse distribution when stock is short
5. Report families yet to collect and stock remaining

Stretch Features (optional)

- Allow partial collection with the balance carried within the month
- Flag unusual patterns, such as many collections in a few minutes

Interview talking points: Entitlement rules, preventing duplicates, stock reconciliation.

CP-078 GasGo: LPG Refill Booking and Delivery Management System

Level: Foundation **Domain:** Public Services, Utilities and Smart City

Gas agencies receive bookings by phone and must enforce a minimum gap between refills.

Core Features (all 5 compulsory)

1. Register consumers with connection type (single or double cylinder)
2. Book a refill only if the minimum gap since the last delivery has passed
3. Track delivery status
4. Track agency stock of filled cylinders
5. List pending bookings older than 3 days

Stretch Features (optional)

- Apply subsidy flags and show subsidy amount
- Plan deliveries by area

Interview talking points: Date-gap rules, stock tracking, pending lists.

CP-079 VoteSecure: Polling Booth Verification and Vote Counting Simulator

Level: Intermediate **Domain:** Public Services, Utilities and Smart City

Election officials at a booth must verify voters, prevent double voting and report turnout through the day.

Core Features (all 5 compulsory)

1. Load the voter list for a booth
2. Verify a voter by voter id
3. Mark a voter as voted and block a second attempt
4. Show hourly turnout percentage
5. Count votes per candidate and declare the result, handling a tie

Stretch Features (optional)

- Show turnout by age group and gender
- Flag voter ids not found in the list

Interview talking points: Verification and duplicate prevention, time-based reporting, counting.

CP-080 TaxAssess: Multi-Factor Property Tax Computation Engine

Level: Foundation **Domain:** Public Services, Utilities and Smart City

Municipal property tax depends on area, zone, use and age of the building, and citizens rarely understand the calculation.

Core Features (all 5 compulsory)

1. Store zone rates and usage factors (residential, commercial)
2. Calculate tax from built-up area, zone, usage and building age with depreciation
3. Apply a rebate for payment before a due date
4. Apply a penalty for late payment
5. Print a receipt with the full calculation

Stretch Features (optional)

- Calculate arrears for unpaid years
- Report collection by zone

Interview talking points: Multi-factor formulas, rebates and penalties, explaining a calculation.

Media, Sports and Entertainment

CP-081 CricStats: Ball-by-Ball Cricket Scoring and Analytics Engine

Level: Intermediate **Domain:** Media, Sports and Entertainment

Local and college cricket matches are scored on paper and statistics are lost.

Core Features (all 5 compulsory)

1. Set up two teams and their batting order
2. Enter the match ball by ball: runs, extras and wickets
3. Maintain batting and bowling figures automatically
4. Show run rate and, in the second innings, the required run rate
5. Print the full scorecard with fall of wickets

Stretch Features (optional)

- Show the partnership for every wicket
- Pick the player of the match by a points formula

Interview talking points: Event-driven state updates, derived statistics, formatted scorecards.

CP-082 FantasyXI: Constraint-Validated Fantasy Team Builder and Points Engine

Level: Advanced **Domain:** Media, Sports and Entertainment

Fantasy sports apps have strict team rules and a points system. This project implements both.

Core Features (all 5 compulsory)

1. Store a player pool with role (batter, bowler, all-rounder, keeper), team and credits
2. Build a team of 11 within 100 credits, with minimum players per role and at most 7 from one side
3. Reject an invalid team with the exact rule broken
4. Calculate fantasy points from match statistics with captain double points
5. Show a leaderboard of users

Stretch Features (optional)

- Suggest the best possible team within the credit limit from past points
- Allow team edits only before the match start time

Interview talking points: Constraint validation, scoring systems, optimisation under a budget.

CP-083 ScreenSeat: Cinema Show and Seat Booking System

Level: Foundation **Domain:** Media, Sports and Entertainment

A single-screen or small multiplex wants its own booking system instead of paying high commissions.

Core Features (all 5 compulsory)

1. Set up screens, shows and seat layout with categories (classic, premium)
2. Show the seat map for a show
3. Book seats and print a ticket
4. Cancel a booking with a refund rule
5. Show occupancy and revenue per show and per movie

Stretch Features (optional)

- Food combo add-ons at booking
- Hold seats for 5 minutes before confirming

Interview talking points: Seat maps, bookings and cancellations, revenue reporting.

CP-084 BeatQueue: Playlist Management and Smart Shuffle Engine

Level: Foundation **Domain:** Media, Sports and Entertainment

Music apps manage songs, playlists and play queues. This project builds that core.

Core Features (all 5 compulsory)

1. Maintain a song library with title, artist, genre and duration
2. Create playlists and add or remove songs
3. Play a playlist in order, with next and previous
4. Shuffle a playlist so no song repeats before all have played
5. Show total duration and the most played songs

Stretch Features (optional)

- Smart playlist of a genre under a chosen total duration
- Remove duplicate songs across playlists

Interview talking points: List manipulation, shuffling without repetition, play counts.

CP-085 FixtureGen: Round-Robin Tournament Scheduling Engine

Level: Intermediate **Domain:** Media, Sports and Entertainment

Organisers of college tournaments make fixtures by hand and teams end up with uneven schedules.

Core Features (all 5 compulsory)

1. Register teams
2. Generate round-robin fixtures so every team plays every other team once
3. Record results
4. Maintain a points table with wins, losses, draws and points
5. Break ties by goal or run difference

Stretch Features (optional)

- Generate a knockout bracket from the top teams
- Avoid any team playing on consecutive days

Interview talking points: Round-robin scheduling algorithm, standings, tie-breaking.

CP-086 FitTrack: Gym Membership and Access Management System

Level: Foundation **Domain:** Media, Sports and Entertainment

A gym owner loses revenue because renewals are missed and attendance is not tracked.

Core Features (all 5 compulsory)

1. Store membership plans with duration and price
2. Register members with plan and start date
3. List members whose plan expires within 7 days
4. Record daily check-ins and block expired members
5. Show monthly revenue and active members

Stretch Features (optional)

- Assign trainers and show each trainer's members
- Freeze a membership for up to 30 days

Interview talking points: Date-based expiry, access rules, business reporting.

CP-087 WatchNext: Genre-Preference Content Recommendation Engine

Level: Intermediate **Domain:** Media, Sports and Entertainment

Streaming platforms suggest titles based on what you watched and rated. This is a transparent, rule-based version.

Core Features (all 5 compulsory)

1. Store titles with genres and language
2. Record a user's watched titles with ratings
3. Calculate a genre preference score for the user from ratings
4. Recommend 5 unwatched titles that best match the preference score
5. Show trending titles by number of views in the last 7 days

Stretch Features (optional)

- Explain each recommendation, for example "because you rated thrillers highly"
- Filter recommendations by language

Interview talking points: Preference scoring, ranking, explainable recommendations.

CP-088 AthleteIQ: Athlete Performance Tracking and Selection System

Level: Intermediate **Domain:** Media, Sports and Entertainment

Coaches test athletes regularly but cannot see who is improving and who is not.

Core Features (all 5 compulsory)

1. Register athletes with sport and age group
2. Record test results (for example sprint time, jump distance) by date
3. Show each athlete's improvement percentage between first and latest test
4. Rank athletes per test within an age group
5. Produce a selection list using minimum standards

Stretch Features (optional)

- Flag athletes whose performance dropped in two consecutive tests
- Show a text chart of progress for an athlete

Interview talking points: Time-series records, improvement metrics, ranking and selection.

CP-089 PostPilot: Multi-Platform Social Media Scheduling Engine

Level: Intermediate **Domain:** Media, Sports and Entertainment

Small businesses post on several platforms and forget, or post at bad times.

Core Features (all 5 compulsory)

1. Create a post with text, platform and scheduled time
2. Check the character limit for each platform
3. Keep a minimum gap between two posts on the same platform
4. Show the posting calendar for the week
5. Mark posts as published and list missed posts

Stretch Features (optional)

- Count hashtags and warn when there are too many
- Suggest the best posting time from past engagement entered by the user

Interview talking points: Validation by platform rules, scheduling, calendars.

CP-090 TuneRecap: Music Listening Analytics and Year-in-Review Generator

Level: Intermediate **Domain:** Media, Sports and Entertainment

Year-end music summaries are hugely popular. This project builds one from a listening log.

Core Features (all 5 compulsory)

1. Load a listening log: song, artist, genre, date and minutes played
2. Show the top 5 songs, artists and genres
3. Show total listening time and the busiest month
4. Show listening by hour of day as a text chart
5. Find the longest listening streak in consecutive days

Stretch Features (optional)

- Compare two users and show a taste-match percentage
- Export the summary as a shareable text card

Interview talking points: Aggregation, ranking, streak detection, presenting insights.

Productivity, Security and Developer Tools

CP-091 CipherVault: Password Strength Analyser and Credential Vault

Level: Intermediate **Domain:** Productivity, Security and Developer Tools

People reuse weak passwords everywhere. This tool checks strength and stores credentials behind a master password.

Core Features (all 5 compulsory)

1. Score a password on length, character variety and common patterns, and show why it is weak
2. Suggest a stronger password
3. Protect the vault with a master password
4. Store site, username and password in encoded form (a simple cipher, clearly labelled as a learning implementation)
5. Search the vault and flag reused passwords

Stretch Features (optional)

- Lock the vault after 3 wrong master passwords
- Check passwords against a list of common passwords

Interview talking points: Validation rules, basic encoding, and being honest about real-world security limits.

CP-092 PriorityPro: Eisenhower-Matrix Task Prioritisation Engine

Level: Foundation **Domain:** Productivity, Security and Developer Tools

Students and professionals keep long to-do lists but do not know what to do first.

Core Features (all 5 compulsory)

1. Add tasks with title, due date, importance and effort
2. Classify tasks into the four quadrants (urgent and important, and so on)
3. Show tasks in order of priority
4. Mark tasks done and list overdue tasks
5. Show weekly completed tasks and on-time percentage

Stretch Features (optional)

- Recurring tasks
- Suggest a plan for today within a given number of hours

Interview talking points: Classification rules, sorting by several keys, productivity metrics.

CP-093 LogLens: Server Log Analytics and Anomaly Detection Tool

Level: Intermediate **Domain:** Productivity, Security and Developer Tools

When a website slows down, engineers dig through thousands of log lines. An analyzer surfaces the problem in seconds.

Core Features (all 5 compulsory)

1. Read log lines with time, IP, endpoint, status code and response time
2. Count requests by status code
3. Show the top 5 IPs and endpoints by number of requests
4. Show errors per hour and the slowest endpoints by average response time
5. Flag suspicious IPs with many failed logins (status 401)

Stretch Features (optional)

- Filter the analysis to a time window
- Write a summary report file

Interview talking points: Parsing text, grouping and counting, finding anomalies. Very relevant for support and DevOps roles.

CP-094 CopyCheck: Text Similarity and Plagiarism Detection Engine

Level: Advanced **Domain:** Productivity, Security and Developer Tools

Colleges need a quick first check on whether two assignments are suspiciously similar.

Core Features (all 5 compulsory)

1. Read two or more text inputs
2. Normalise the text: lowercase, remove punctuation and common words
3. Calculate word-frequency similarity between each pair
4. List sentences that appear in both texts
5. Print a similarity report with a percentage per pair

Stretch Features (optional)

- Compare a set of 10 submissions and rank the most similar pairs
- Detect similarity using 3-word sequences instead of single words

Interview talking points: String processing, similarity measures, pairwise comparison.

CP-095 BugBoard: Defect Lifecycle Tracking System

Level: Intermediate **Domain:** Productivity, Security and Developer Tools

Small development teams track bugs in chat messages and lose them.

Core Features (all 5 compulsory)

1. Log a bug with title, module, severity and steps
2. Assign it to a developer
3. Move it through New, Assigned, Fixed, Verified, Closed or Reopened, allowing only valid steps
4. Count how many times each bug was reopened
5. Report open bugs by age and the module with the most bugs

Stretch Features (optional)

- Mark duplicates and link them to the original
- Show each developer's fix rate

Interview talking points: Workflow states, metrics, the actual process used in software teams.

CP-096 DiffTrack: Document Versioning and Diff Engine

Level: Advanced **Domain:** Productivity, Security and Developer Tools

Writers and developers need to see what changed between two versions of a document and to roll back mistakes.

Core Features (all 5 compulsory)

1. Save versions of a text document with author and time
2. Show the list of versions
3. Show a line-by-line difference between any two versions (added, removed, unchanged)
4. Roll back to an earlier version as a new version
5. Show who changed the most lines

Stretch Features (optional)

- Use the longest common subsequence method for a better difference
- Show a word-level difference for changed lines

Interview talking points: Comparing sequences, the idea behind version control systems such as Git.

CP-097 ShortLink: Base62 URL Shortening and Click Analytics Service

Level: Intermediate **Domain:** Productivity, Security and Developer Tools

Link shorteners turn long URLs into short codes and count clicks. This project builds the core logic.

Core Features (all 5 compulsory)

1. Generate a short code for a long URL using a counter converted to base 62
2. Expand a short code back to the original URL
3. Return the same code if the same long URL is shortened again
4. Count clicks per code
5. Allow custom aliases, validated for allowed characters and uniqueness

Stretch Features (optional)

- Expiry date for links
- Report the top links by clicks

Interview talking points: Encoding numbers in base 62, two-way lookup, a popular system design interview topic.

CP-098 FocusFlow: Study Session Analytics and Streak Tracker

Level: Foundation **Domain:** Productivity, Security and Developer Tools

Students preparing for placements need to see whether their study time is consistent.

Core Features (all 5 compulsory)

1. Log study sessions with subject, start and end time
2. Show daily focus minutes
3. Show time per subject for the week
4. Calculate the current and longest daily study streak
5. Set a weekly goal and show progress

Stretch Features (optional)

- Pomodoro mode: 25-minute sessions with break reminders counted
- Monthly report with best and worst week

Interview talking points: Time calculations, streaks, goal tracking.

CP-099 CleanSlate: Data Validation and Cleansing Utility

Level: Intermediate **Domain:** Productivity, Security and Developer Tools

Companies spend a large share of data work cleaning bad records before they can use them. This utility automates the common checks.

Core Features (all 5 compulsory)

1. Read records with name, email, phone, date of birth and city
2. Detect invalid emails, phone numbers and dates
3. Detect duplicate records
4. Suggest fixes such as trimming spaces and standardising case
5. Write the cleaned records and an error report listing every problem found

Stretch Features (optional)

- Let the user define validation rules for a new column
- Show a data quality score for the file

Interview talking points: Validation logic, string processing, the reality of working with data.

CP-100 AskCampus: Rule-Based College FAQ Chatbot

Level: Intermediate **Domain:** Productivity, Security and Developer Tools

The college office answers the same questions every day: fees, exam dates, hostel rules. A simple chatbot can handle most of them.

Core Features (all 5 compulsory)

1. Maintain a knowledge base of questions, answers and keywords
2. Match a user's question to the best answer by keyword overlap
3. Reply with a fallback message when no answer scores high enough
4. Log unanswered questions
5. Let an admin add new questions and answers

Stretch Features (optional)

- Handle simple spelling mistakes in keywords
- Show the most asked questions

Interview talking points: Text matching and scoring, fallback design, and how this differs from AI chatbots.

Template 1: Project Plan Card

Fill this in the launch session. The trainer will check it with you during the session. Keep it; in the review, the trainer may test your program against the "What can go wrong" column.

Item	Your answer
Problem ID and title	
Team members, USNs, track	
Who uses the program	
Data stored for one record (list every field)	
Menu options	

No.	Core Feature in your own words	What can go wrong (at least 2)	Week
1			
2			
3			
4			
5			

Template 2: README

Your repository must contain a file named README.md with these five sections.

```
# CP-017 TrackPoint: Order Lifecycle and Delivery Tracking System
## Team
Ravi K (1AT23CS045), Asha M (1AT23CS051), Java track
## How to run
javac Main.java
java Main
## Features that work
1. Place an order ...   (list all Core and Stretch Features that work)
## Sample run
(paste console output or add a screenshot)
```

Template 3: Resume Line and 60-Second Pitch

You will write these at the end of the review session. Prepare a draft before you come.

Resume line

Pattern: **Built a** (what) **in** (language) **that** (what it does), **handling** (what was hard).

> Example: Built TrackPoint, an order lifecycle and delivery tracking system in Java that moves orders through 5 delivery stages, blocks invalid status changes and flags shipments delayed for more than 2 days.

60-second pitch (4 sentences)

1. The problem it solves, and for whom.
2. What you built: the main features.
3. The hardest part, and how you solved it.
4. What you would add next.

Where to use it

- Resume: under Projects, with the GitHub link.
- LinkedIn: in a post (Template 4) and under Projects on your profile.
- Interviews: say the pitch out loud at least three times before your first interview.

Template 4: LinkedIn Post

You will publish this post in the review session. Recruiters read LinkedIn. A good project post is one of the easiest ways to get noticed.

Before the review session

- If you do not have a LinkedIn account, create one. Add a clear photo and a headline, for example: "Computer Science student at [college] | Java | Building real-world projects".
- Install the LinkedIn app on your phone and make sure you can log in.

Step by step, in the session

1. Open LinkedIn (app or linkedin.com) and log in.
2. Open ChatGPT or Claude and paste the prompt below, with your own details filled in.
3. Read what the AI writes. Remove anything that is not true about your project. Keep it under 200 words.
4. In LinkedIn, tap **Start a post** and paste your edited text.
5. Add your GitHub link at the end.
6. Tag the people and pages your trainer gives you (your college's placement head, your college page and your CDC page). Type @ followed by the name and pick the right one from the list.
7. Add a screenshot of your program running, if you can.
8. Tap **Post**. Copy the post link and share it with your trainer.

Prompt to give the AI

Write a LinkedIn post of 120 to 180 words announcing my capstone project.

Project name: [name from the problem bank]

What it does: [2 lines in your own words]

Language: [Java / Python / C]

The hardest problem I solved: [1 line]

What I learnt: [1 line]

Tone: confident, energetic and professional, the way a young engineer would announce a project. Use strong action words.

End with 4 to 6 relevant hashtags.

Do not exaggerate, and do not add any feature I have not built.

Sample post

Excited to share my capstone project: TrackPoint, an Order Lifecycle and Delivery Tracking System built in Java.

Every online order moves through a chain of stages, from Placed to Delivered. TrackPoint models that chain as a state machine, blocks invalid status jumps, keeps a full audit trail for every order and flags shipments stuck for more than 2 days.

The toughest part? Designing the transition rules so that no order can ever skip a step, and handling every kind of invalid input without a single crash.

Built as part of the Coding for Placement programme at [college]. Grateful to [placement head] and [CDC page] for the opportunity.

Code: [GitHub link]

#Java #SoftwareEngineering #CapstoneProject #ProblemSolving #LearningByBuilding